FW-H equation
Far-field approximation for impenetrable surfaces

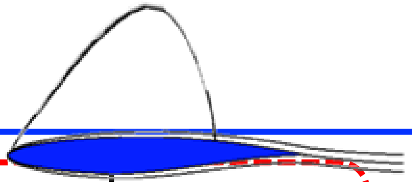
- **Far-field approximation of the FW-H equation for *impenetrable* surfaces**:
  - Flow velocity normal to the surface equals the surface normal velocity - $v_i n_i = V_i n_i$

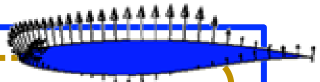
$\rho - \rho' = \rho_\infty$

$$\rho'(\mathbf{x},t)c_\infty^2 \approx \frac{x_i x_j}{|\mathbf{x}|^2}\frac{1}{c_\infty^2}\frac{\partial^2}{\partial t^2}\int_{V_o}\left[\frac{T_{ij}}{4\pi|\mathbf{x}||1-M_r|}\right]_{\tau=\tau^*} dV(\mathbf{z}) + \frac{x_i}{|\mathbf{x}|}\frac{1}{c_\infty}\frac{\partial}{\partial t}\int_{S_o}\left[\frac{p_{ij}n_j}{4\pi|\mathbf{x}||1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z})$$
$$+\frac{\partial}{\partial t}\int_{S_o}\left[\frac{\rho_\infty V_j n_j}{4\pi|\mathbf{x}||1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z})$$

FW-H equation

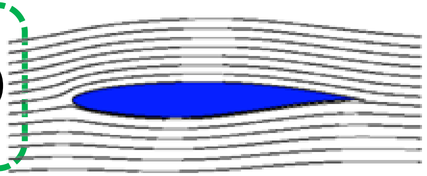
Application for rotors/propellers





$$\rho'(\mathbf{x},t)c_\infty^2 \approx \frac{x_i x_j}{|\mathbf{x}|^2}\frac{1}{c_\infty^2}\frac{\partial^2}{\partial t^2}\int_{V_o}\left[\frac{T_{ij}}{4\pi|\mathbf{x}||1-M_r|}\right]_{\tau=\tau^*} dV(\mathbf{z}) + \frac{x_i}{|\mathbf{x}|}\frac{1}{c_\infty}\frac{\partial}{\partial t}\int_{S_o}\left[\frac{p_{ij}n_j}{4\pi|\mathbf{x}||1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z})$$

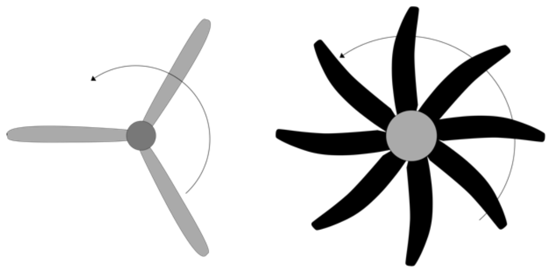
$$+\frac{\partial}{\partial t}\int_{S_o}\left[\frac{\rho_\infty V_j n_j}{4\pi|\mathbf{x}||1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z})$$

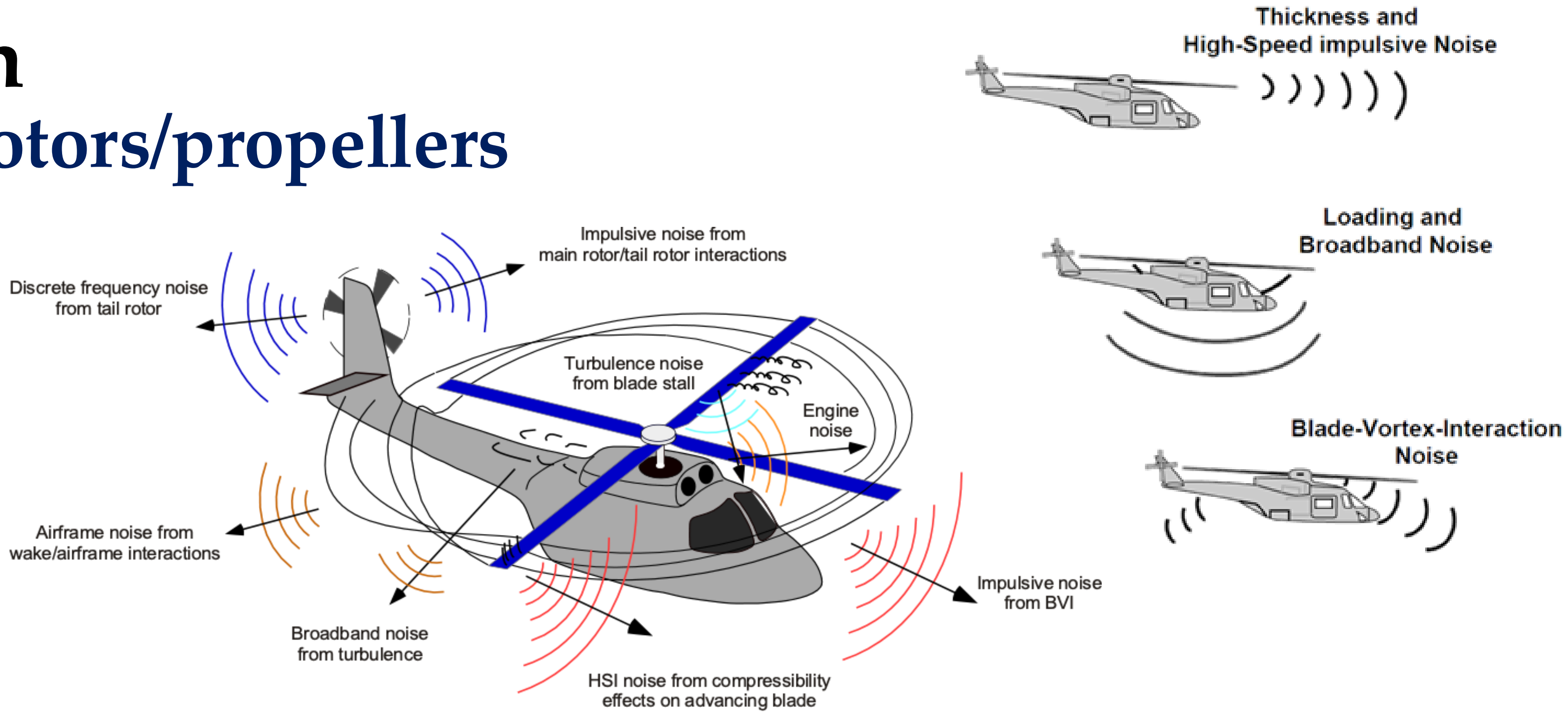


- Three main noise sources are recognized for *rotating blades* **with impenetrable blades:**
  - **Quadrupole noise** (T_{ij}) – due to **turbulence** and **flow distortions** (e.g., shock waves)
  - **Loading (dipole) noise** ($p_{ij}n_j$) – due to **surface loading (force)**
  - **Thickness (monopole) noise** ($\rho_\infty V_j n_j$) – due to a **volume displacement** source arising when the observer "sees" a time varying surface velocity at emission time
    - Zero for an object in linear motion traveling towards the observer, but non-zero for a rotating blade that continuously changes direction relative to the observer

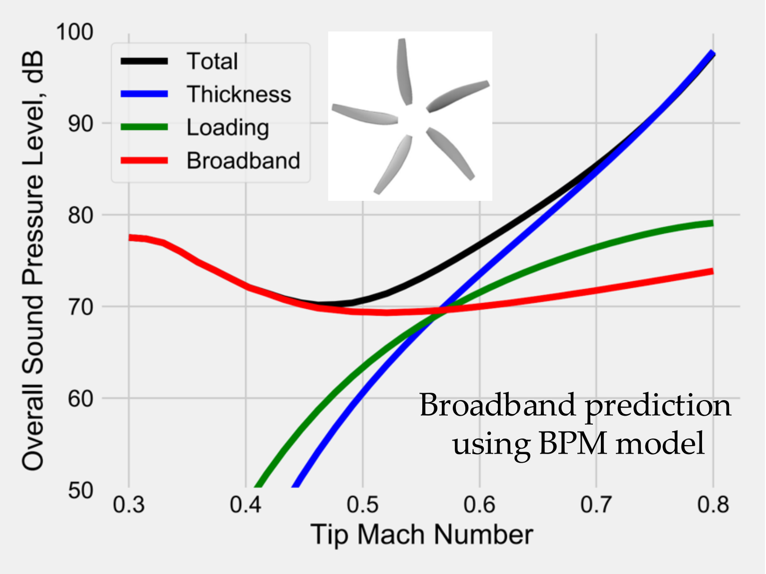
FW-H equation
Application for rotors/propellers

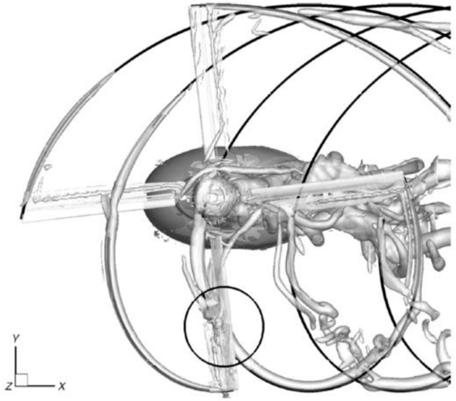
1950's　　Modern

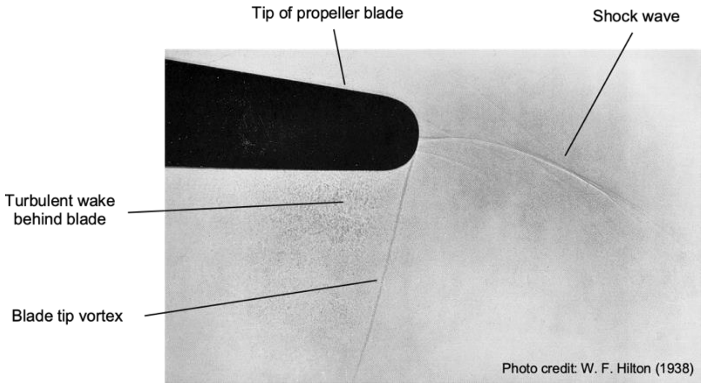




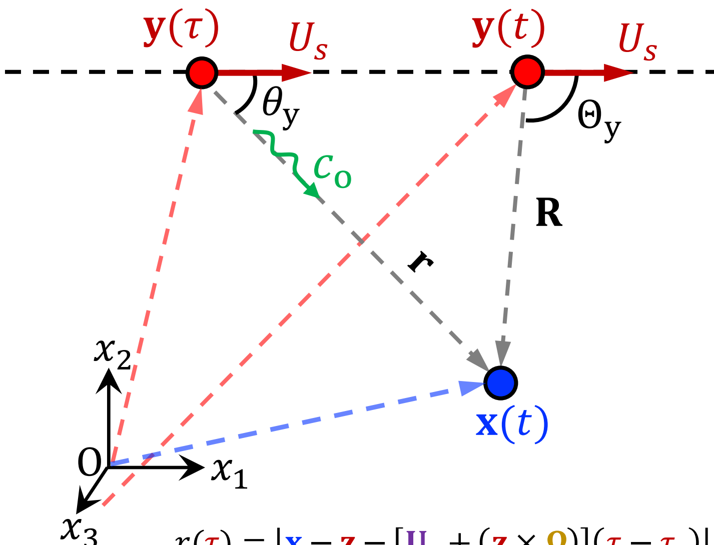
eVTOL rotor FW-H prediction
(thickness + loading)







Moving sources



$r(\tau) = |\mathbf{x} - \mathbf{z} - [\mathbf{U}_s + (\mathbf{z} \times \boldsymbol{\Omega})](\tau - \tau_o)|$

- Consider an **acoustically compact source** moving at **constant speed** U_s
  - Ignore sound from turbulence ($T_{ij} \approx 0$) and assume stationary medium
- Distance to observer is given by: $r(\tau) = \left|\mathbf{x} - \mathbf{y}^{(c)}\right| = |\mathbf{x} - U_s\tau|$
  - $\mathbf{y}^{(c)}$ - centroid of the surface, at the correct retarded time, in the fixed reference frame, corresponding to $\mathbf{z} = 0$ in the moving frame
  - We take the time origin of the surface motion $\tau_0 = 0$
- Applying FW-H equation for an impenetrable surface:

$$\rho'(\mathbf{x},t)c_\infty^2 \approx -\frac{\partial}{\partial x_i}\int_{S_o}\left[\frac{p_{ij}n_j}{4\pi r|1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z}) + \frac{\partial}{\partial t}\int_{S_o}\left[\frac{\rho_\infty V_j n_j}{4\pi r|1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z})$$

  - But since the source is acoustically compact, we can take the propagation factor outside of the integrals:

$$\rho'(\mathbf{x},t)c_\infty^2 \approx -\frac{\partial}{\partial x_i}\left[\frac{1}{4\pi r|1-M_r|}\int_{S_o} p_{ij}n_j dS(\mathbf{z})\right]_{\tau=\tau^*} + \frac{\partial}{\partial t}\left[\frac{1}{4\pi r|1-M_r|}\cancel{\int_{S_o} \rho_\infty V_j n_j dS(\mathbf{z})}\right]_{\tau=\tau^*}$$

=0 for any closed rigid body

Moving sources

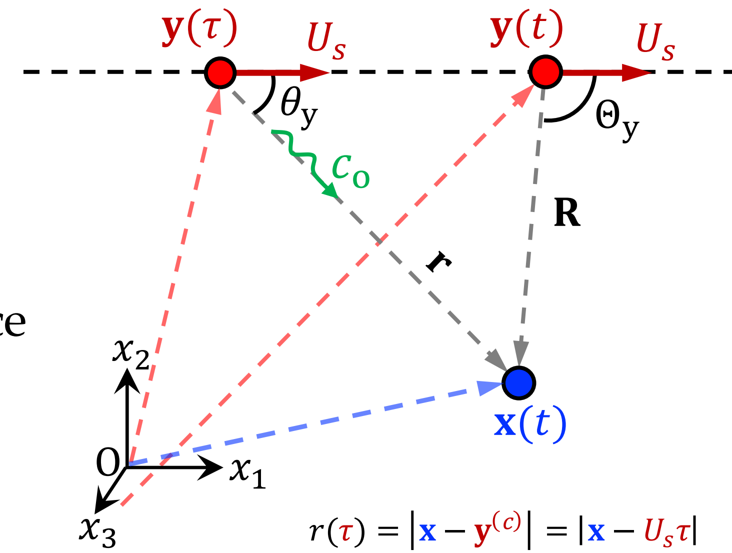
- Thus, only **surface loading** will radiate sound
- For modelling, assume a harmonic dipole source in the frame of reference moving with the surface

  $\hat{F}_i$ - vector force amplitude

$$\int_{S_o} p_{ij} n_j dS(\mathbf{z}) = \mathcal{R}\mathrm{e}\left(\hat{F}_i e^{-i\omega_o \tau}\right)$$

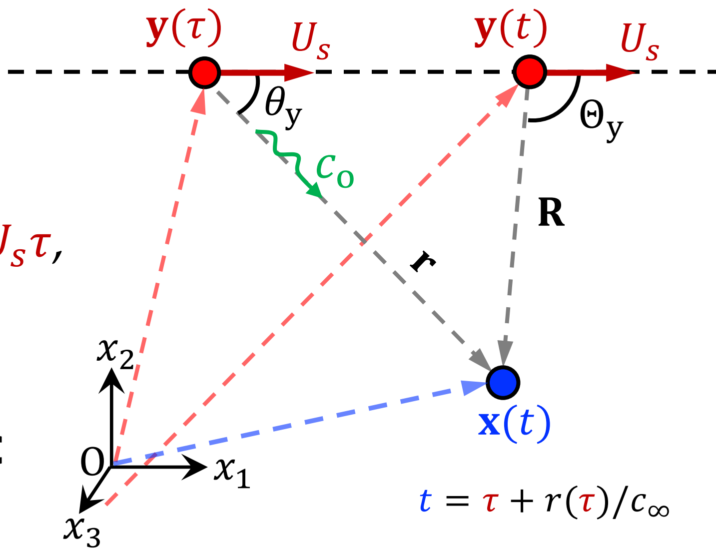
- The **acoustic field** is therefore:

$$\boxed{\rho'(\mathbf{x}, t) c_\infty^2 \approx -\mathcal{R}\mathrm{e}\left(\frac{\partial}{\partial x_i}\left[\frac{\hat{F}_i e^{-i\omega_o \tau}}{4\pi r|1 - M_r|}\right]_{\tau=\tau^*}\right)}$$



$$r(\tau) = |\mathbf{x} - \mathbf{y}^{(c)}| = |\mathbf{x} - U_s \tau|$$

Moving sources



- In the **far-field**, retarded time equation can be approximated with $y_1 = U_s\tau$, yielding:

$$r(\tau) \approx |\mathbf{x}| - \frac{x_1 U_s \tau}{|\mathbf{x}|} + \cdots$$

  - Approximation only applies over short periods ($U_s \ll |\mathbf{x}|$), such that $x_1/|\mathbf{x}| = \mathbb{C}$

- **Velocity** and **Mach number** of the **source** in the direction of the observer:

$$U_r = \frac{U_s x_1}{|\mathbf{x}|} \qquad M_r = \frac{U_r}{c_\infty}$$

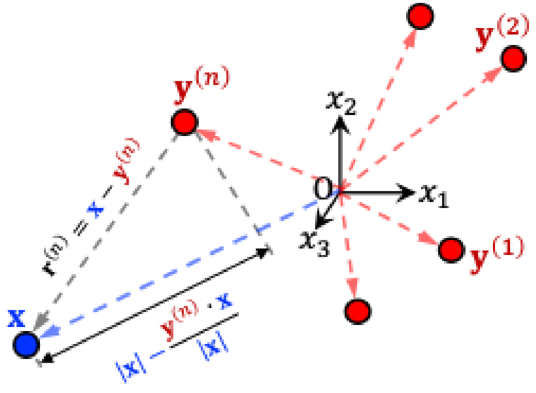
- The far field approximation applied then gives: $t - \tau \approx |\mathbf{x}|/c_\infty - M_r\tau$ or $\tau \approx (t - |\mathbf{x}|/c_\infty)/(1 - M_r)$
- Therefore, the **far-field** approximation gives:

Ignoring near field terms of order $|\mathbf{x}|^{-2}$

$$\Longrightarrow \rho'(\mathbf{x},t)c_\infty^2 \approx -\mathcal{R}e\left(\frac{\partial}{\partial x_i}\left[\frac{\hat{F}_i e^{-i\omega_o(t-|\mathbf{x}|/c_\infty)/(1-M_r)}}{4\pi|\mathbf{x}||1-M_r|}\right]\right)$$

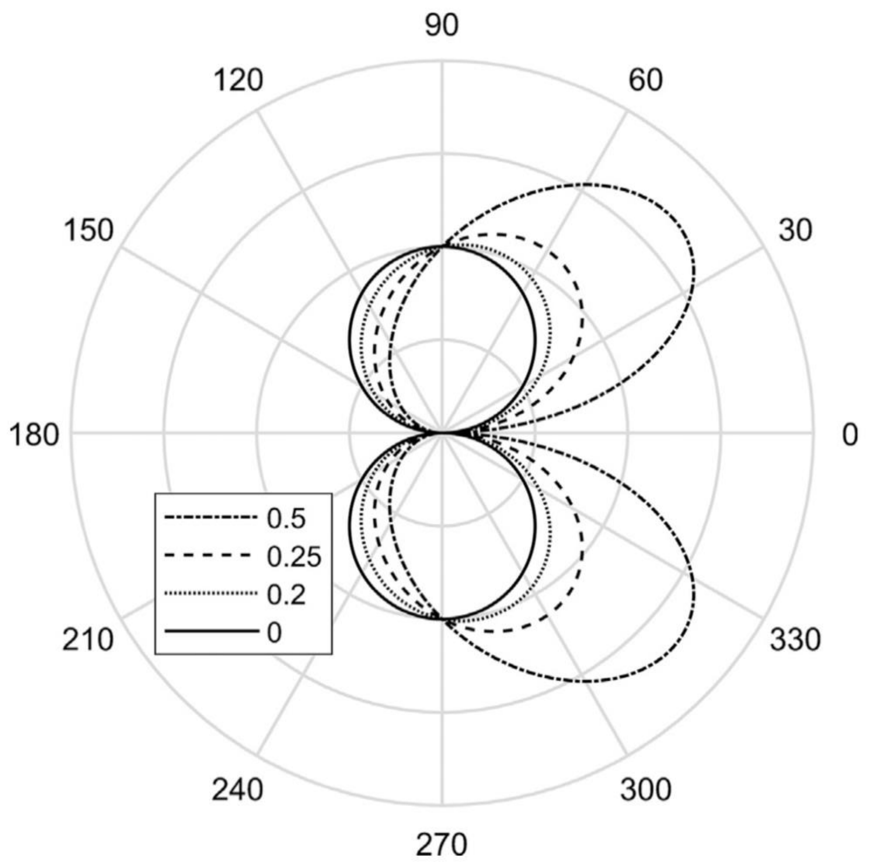
$$\boxed{\rho'(\mathbf{x},t)c_\infty^2 \approx \mathcal{R}e\left(\frac{-i\omega_o x_i \hat{F}_i e^{-i\omega_o(t-|\mathbf{x}|/c_\infty)/(1-M_r)}}{4\pi c_\infty|\mathbf{x}|^2(1-M_r)^2} + \mathcal{O}(|\mathbf{x}|^{-2})\right)}$$

Doppler frequency shift



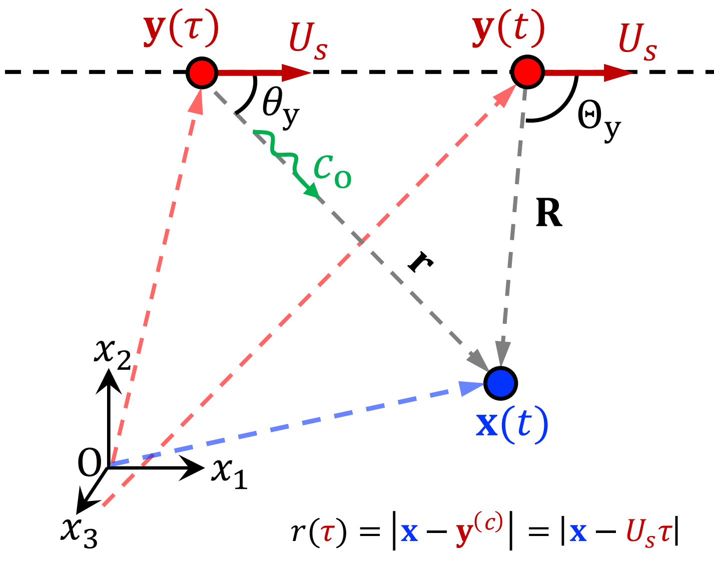
$$r = |\mathbf{x} - \mathbf{y}| \approx |\mathbf{x}| - \frac{\mathbf{y}\cdot\mathbf{x}}{|\mathbf{x}|}$$

Moving sources



Directivity of a dipole source with an axis perpendicular to its direction of motion as a function of source Mach number

Moving sources



$r(\tau) = |\mathbf{x} - \mathbf{y}^{(c)}| = |\mathbf{x} - U_s \tau|$

- Retarded time is evaluated by solving $t = \tau + r(\tau)/c_\infty$, or:

$$t - \tau = \left((x_1 - U_s\tau)^2 + x_2^2 + x_3^2\right)^{1/2} / c_\infty$$

- Retarded time $\tau = \tau^*$ can be obtained in terms of the observer time by squaring the above equation:

$$c_\infty^2 (t - \tau)^2 = (x_1 - U_s\tau)^2 + x_2^2 + x_3^2$$

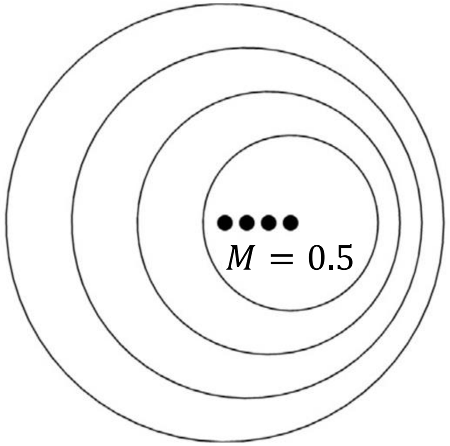
$$\boxed{\tau^* = \frac{c_\infty t - M x_1 \pm \sqrt{(x_1 - U_s t)^2 + (1 - M^2)(x_2^2 + x_3^2)}}{c_\infty (1 - M^2)}}$$

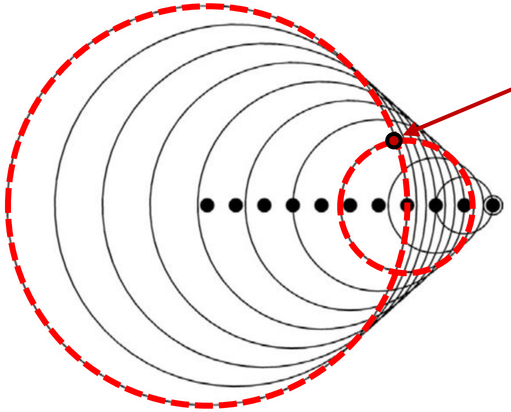
$$M = \frac{U_s}{c_\infty}$$

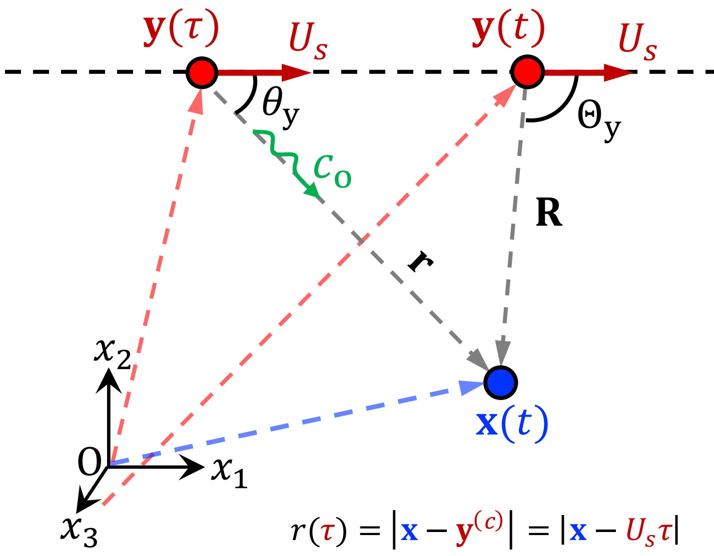
- **Two possible solutions**: τ_1 $(-)$ and τ_2 $(+)$ | $\tau_2 > \tau_1$
  - Solution is constrained by **causality condition**: $t > \tau$
- <u>**For subsonic flow**</u> $(M < 1)$ $\rightarrow$ **only** τ_1 **applies** $(-)$
- <u>**For supersonic flow**</u> $(M > 1)$ $\rightarrow$ **two *real* solutions** are possible if: $(x_1 - U_s t)^2 > (M^2 - 1)(x_2^2 + x_3^2)$

Moving sources

Wave fronts from a source moving from left to right

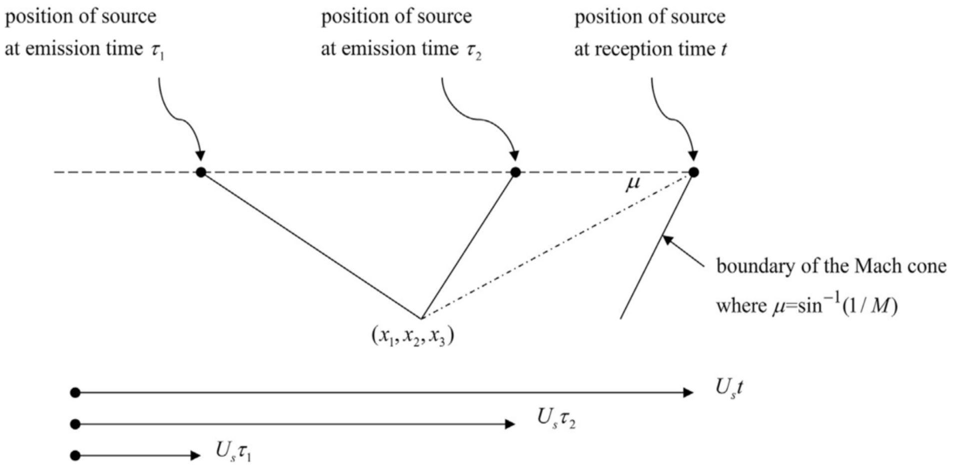






$r(\tau) = |\mathbf{x} - \mathbf{y}^{(c)}| = |\mathbf{x} - U_s\tau|$

$M = \dfrac{U_s}{c_\infty}$



$$(x_1 - U_s t)^2 > (M^2 - 1)(x_2^2 + x_3^2)$$

$$\Rightarrow \frac{|x_1 - U_s t|}{\sqrt{x_2^2 + x_3^2}} = \cot\mu > \sqrt{M^2 - 1}$$

$$\boxed{\mu < \sin^{-1}\left(\frac{1}{M}\right)}$$ **Mach cone angle**

μ – angle of the observer to the path of the source evaluated at reception time

Sources in a free stream

- In many applications, we are interested in the sound radiation from a uniform flow over a stationary object (CFD, WT, propeller WTT, …)
- How to deal with sound propagation in a **uniform free stream**?

> Recall that the presence of uniform flow **contradicts** the assumption of a quiescent medium made in both Lighthill's and FW-H analogies

- To solve such cases - **work in the reference frame that moves with the fluid**

$$\boxed{\mathbf{v} = \mathbf{w} + \mathbf{U}^{(\infty)}}$$

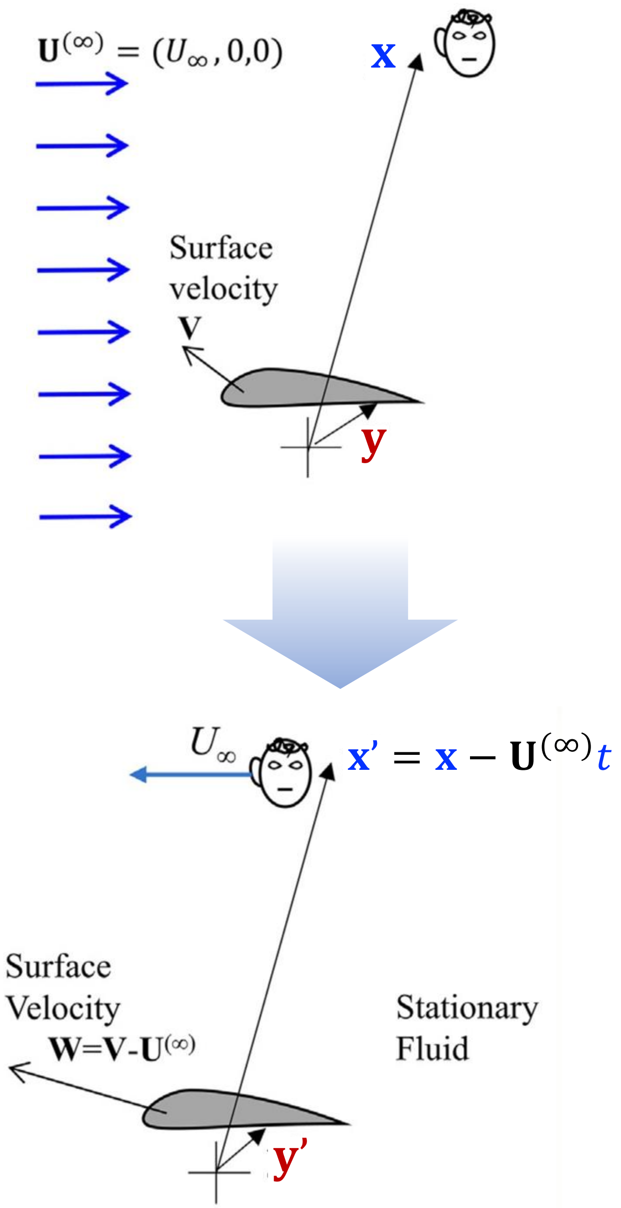
<u>Flow velocity</u> in *fixed* frame

<u>Flow velocity</u> in *moving* frame

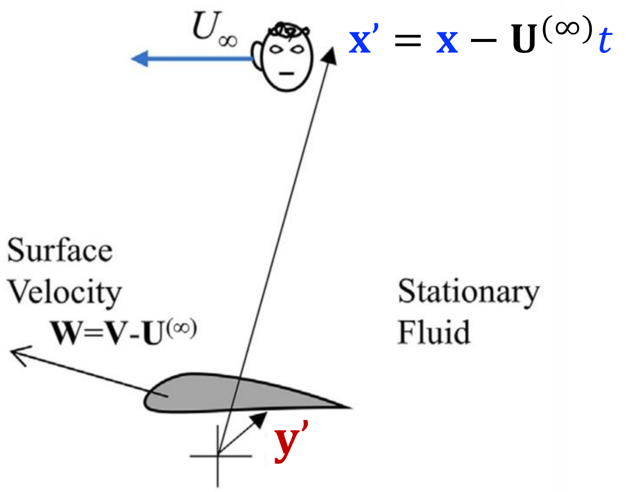
$$\boxed{\mathbf{V} = \mathbf{W} + \mathbf{U}^{(\infty)}}$$

<u>Surface velocity</u> in *fixed* frame

<u>Surface velocity</u> in *moving* frame



Sources in a free stream



- Consider the solution to Lighthill's wave equation in moving reference
- Lighthill's stress tensor in the moving frame of reference is:

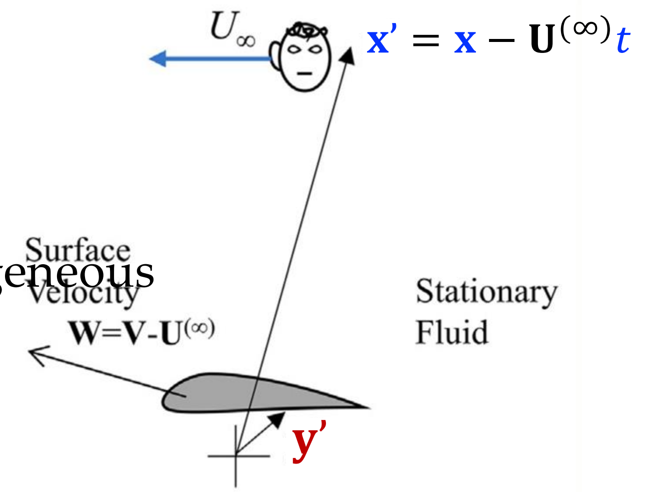
$$T'_{ij} = \rho w_i w_j + [(p - p_\infty) - (\rho - \rho_\infty)c_\infty^2]\delta_{ij} - \sigma_{ij}$$

- Take the previous solution of FW-H equation, but in moving frame of reference:

$$\rho'(\mathbf{x}', t)c_\infty^2 = \int_{-T}^{T}\int_{V_o(\tau)} \frac{\partial^2 G(\mathbf{x}', t|\mathbf{y}', \tau)}{\partial y'_i \partial y'_j} T'_{ij} dV(\mathbf{y}')d\tau + \int_{-T}^{T}\int_{S_o(\tau)} \frac{\partial G(\mathbf{x}', t|\mathbf{y}', \tau)}{\partial y'_i}\left(\left(\rho w_i\left(w_j - W_j\right) + p_{ij}\right)n_j\right) dS(\mathbf{y}')d\tau$$
$$- \int_{-T}^{T}\int_{S_o(\tau)} \frac{\partial G(\mathbf{x}', t|\mathbf{y}', \tau)}{\partial \tau}\left(\left(\rho w_j - \rho' W_j\right)n_j\right) dS(\mathbf{y}')d\tau$$

- To solve the equation, we shall first **change it to be in the fixed points x and y**
  - To do so, we need to *replace the Green's function*

Sources in a free stream



- First, let's deal with the Green's function, which satisfies the inhomogeneous wave equation in a moving frame:

$$\frac{1}{c_o^2}\frac{\partial^2 G}{\partial \tau^2} - \frac{\partial^2 G}{\partial y'^2_i} = \delta(\mathbf{x}' - \mathbf{y}')\delta(t - \tau)$$

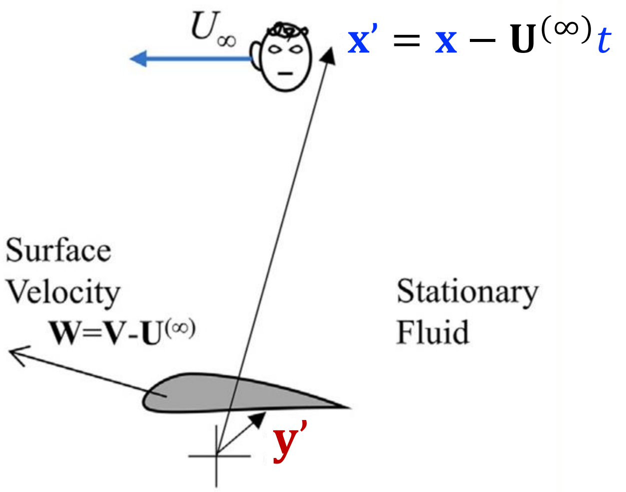
  - Where: $G(\mathbf{x}', t|\mathbf{y}', \tau) = \dfrac{\delta(t - \tau - |\mathbf{x}' - \mathbf{y}'|/c_\infty)}{4\pi|\mathbf{x}' - \mathbf{y}'|}$

- The Green's function in the stationary frame, $G_e(\mathbf{x}, t|\mathbf{y}, \tau)$, is related to G through translation of the coordinates:

$$G_e(\mathbf{x}, t|\mathbf{y}, \tau) = \frac{\delta\left(t - \tau - \left|\left(\mathbf{x} - \mathbf{U}^{(\infty)}t\right) - \left(\mathbf{y} - \mathbf{U}^{(\infty)}\tau\right)\right|/c_\infty\right)}{4\pi\left|\left(\mathbf{x} - \mathbf{U}^{(\infty)}t\right) - \left(\mathbf{y} - \mathbf{U}^{(\infty)}\tau\right)\right|}$$

  - Note that *space derivatives* are unaltered by this transformation: $\dfrac{\partial G}{\partial y'_i} = \dfrac{\partial G_e}{\partial y_i}$
  - *Time derivative* in moving frame is the material derivative in stationary frame: $\dfrac{\partial G}{\partial \tau} = \dfrac{\partial G_e}{\partial \tau} + U_i^{(\infty)}\dfrac{\partial G_e}{\partial y_i} = \dfrac{D_\infty G_e}{D\tau}$

Sources in a free stream



- Applying the former conversions to the inhomogeneous wave equation of the Green's function yields:

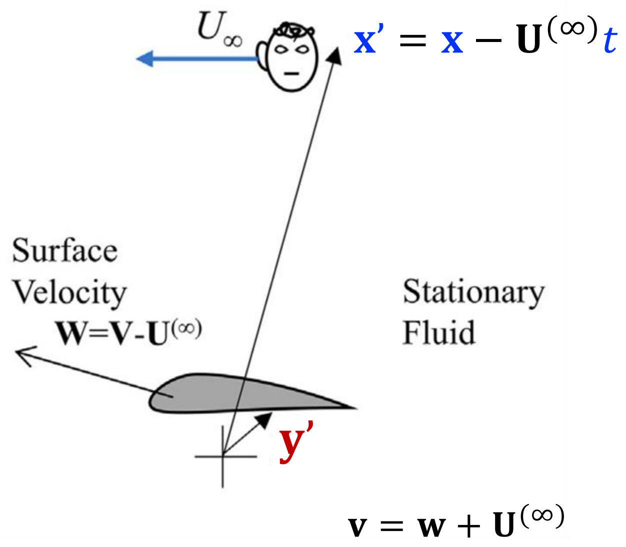
$$\frac{1}{c_o^2}\frac{D_\infty^2 G_e}{D\tau^2} - \frac{\partial^2 G_e}{\partial y_i^2} = \delta(\mathbf{x}-\mathbf{y})\delta(t-\tau)$$

  - Where: $G_e(\mathbf{x},t|\mathbf{y},\tau) = \dfrac{\delta(t-\tau-r_m(t-\tau,\mathbf{x},\mathbf{y})/c_\infty)}{4\pi r_m(t-\tau,\mathbf{x},\mathbf{y})}$ $\qquad r_m(t-\tau,\mathbf{x},\mathbf{y}) = |\mathbf{x}-\mathbf{y}-\mathbf{U}^{(\infty)}(t-\tau)|$

- Thus, after replacing $W_j = V_j - U_j^{(\infty)}$, the solution to FW-H equation becomes:

$$\boxed{\begin{aligned}\rho'(\mathbf{x},t)c_\infty^2 = &\int_{-T}^{T}\int_{V_o(\tau)}\frac{\partial^2 G_e(\mathbf{x},t|\mathbf{y},\tau)}{\partial y_i \partial y_j}T'_{ij}dV(\mathbf{y})d\tau + \int_{-T}^{T}\int_{S_o(\tau)}\frac{\partial G_e(\mathbf{x},t|\mathbf{y},\tau)}{\partial y_i}\left(\left(\rho w_i\left(w_j - V_j + U_j^{(\infty)}\right) + p_{ij}\right)n_j\right)dS(\mathbf{y})d\tau \\ &- \int_{-T}^{T}\int_{S_o(\tau)}\frac{DG_e(\mathbf{x},t|\mathbf{y},\tau)}{D\tau}\left(\left(\rho w_j - \rho'\left(V_j - U_j^{(\infty)}\right)\right)n_j\right)dS(\mathbf{y})d\tau\end{aligned}}$$

Sources in a free stream



- The result is often used for CFD calculations or surface flows in which the **surfaces are stationary**, and the **medium is moving at a uniform velocity**; in such cases we have:
  - $\mathbf{V} = 0$
  - Impermeable BC on the surface requires $\left(\mathbf{U}^{(\infty)} + \mathbf{w}\right) \cdot \mathbf{n} = 0$

$$\left(\rho w_i \left(w_j - \cancel{V_j} + U_j^{(\infty)}\right) + p_{ij}\right) n_j \approx p' n_j$$

$$\left(\rho w_j - \rho' \left(\cancel{V_j} - U_j^{(\infty)}\right)\right) n_j = \underbrace{-\rho_\infty U_j^{(\infty)} n_j}_{\text{Time invariant}}$$

Neglecting σ_{ij}

$$p' = p - p_\infty$$

$$T'_{ij} = \rho w_i w_j + [(p - p_\infty) - (\rho - \rho_\infty) c_\infty^2]\delta_{ij} - \sigma_{ij}$$

$$\boxed{\rho'(\mathbf{x}, t) c_\infty^2 = \int_{-T}^{T} \int_{V_o(\tau)} \frac{\partial^2 G_e(\mathbf{x}, t|\mathbf{y}, \tau)}{\partial y_i \partial y_j} T'_{ij} dV(\mathbf{y}) d\tau + \int_{-T}^{T} \int_{S_o(\tau)} \frac{\partial G_e(\mathbf{x}, t|\mathbf{y}, \tau)}{\partial y_i} p'(\mathbf{y}, \tau) n_j dS(\mathbf{y}) d\tau}$$

- We had this result before, except now Lighthill's stress tensor depends on $\rho w_i w_j$ and **Green's function must satisfy the convective wave equation**

Prandtl-Glauert transformation

- Recall the Green's function we found for sound propagation in a **uniform flow** to be:

$$G_e(\mathbf{x}, t|\mathbf{y}, \tau) = \frac{\delta(t - \tau - r_m(t - \tau, \mathbf{x}, \mathbf{y})/c_\infty)}{4\pi r_m(t - \tau, \mathbf{x}, \mathbf{y})} \qquad r_m(t - \tau, \mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y} - \mathbf{U}^{(\infty)}(t - \tau)|$$

- To simplify the above result, we need to define the correct retarded time τ^* for a source at $\mathbf{y}$ for waves that arrive at the observer $\mathbf{x}$ at reception time t
- We use of the property of Dirac delta functions which applies to any continuous function $g(\tau)$ that is zero when $\tau = \tau^*$:

$$\delta(g(\tau)) = \frac{\delta(\tau - \tau_i^*)}{|\partial g/\partial \tau|} \qquad \frac{\partial g}{\partial \tau} = \frac{\partial}{\partial \tau}(t - \tau - r_m(t - \tau, \mathbf{x}, \mathbf{y})/c_\infty)$$

  - Where: $g = t - \tau - r_m(t - \tau, \mathbf{x}, \mathbf{y})/c_\infty$
- The **correct retarded time** is evaluated from: $\boxed{\tau^* = t - r_m(t - \tau^*, \mathbf{x}, \mathbf{y})/c_\infty}$

Solution (as before…)

$$\boxed{\tau^* = t - \frac{-(x_1 - y_1)M \pm \sqrt{(x_1 - y_1)^2 + (1 - M^2)\left((x_2 - y_2)^2 + (x_3 - y_3)^2\right)}}{c_\infty(1 - M^2)}} \qquad M = \frac{U_\infty}{c_\infty}$$

Prandtl-Glauert transformation

- Let's introduce the variables: $\beta^2 = 1 - M^2$ and $r_g = \sqrt{(x_1 - y_1)^2 + \beta^2\left((x_2 - y_2)^2 + (x_3 - y_3)^2\right)}$

$$\boxed{\tau^* = t - \frac{-(x_1 - y_1)M \pm r_g}{c_\infty \beta^2}}$$

- We shall satisfy the causality condition ($t>\tau^*$): $\boxed{t - \tau^* = \frac{r_m}{c_\infty} = \frac{r_g - (x_1 - y_1)M}{c_\infty \beta^2}} > 0$ $\quad r_g = r_m\beta^2 + (x_1 - y_1)M$

- To complete the definition of the Green's function so it will be in its regular form we evaluate:

$$\left|\frac{\partial g}{\partial \tau}\right| = \left|1 + \frac{1}{c_\infty}\frac{\partial r_m}{\partial \tau}\right| = \left|1 + \frac{1}{c_\infty}\frac{\mathbf{r}_m}{r_m}\cdot\frac{\partial \mathbf{r}_m}{\partial \tau}\right| \Longrightarrow \left|\frac{\partial g}{\partial \tau}\right| = \left|1 + \frac{1}{c_\infty}\frac{\mathbf{r}_m \cdot \mathbf{U}^{(\infty)}}{r_m}\right|$$

$$g = t - \tau - r_m(t-\tau, \mathbf{x}, \mathbf{y})/c_\infty$$
$$r_m(t-\tau, \mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y} - \mathbf{U}^{(\infty)}(t-\tau)|$$
$$\mathbf{r}_m(t-\tau, \mathbf{x}, \mathbf{y}) = \mathbf{x} - \mathbf{y} - \mathbf{U}^{(\infty)}(t-\tau)$$

  - We choose $\mathbf{U}^{(\infty)} = (U_\infty, 0, 0)$, therefore: $\mathbf{r}_m \cdot \mathbf{U}^{(\infty)} = U_\infty(x_1 - y_1) - U_\infty^2(t - \tau^*)$

$$\left|\frac{\partial g}{\partial \tau}\right| = \left|1 + \frac{(x_1 - y_1)M - (t - \tau^*)c_\infty M^2}{r_m}\right|$$

Causality… $\quad t - \tau^* = \frac{r_m}{c_\infty}$

$$\boxed{\left|\frac{\partial g}{\partial \tau}\right| = \left|\frac{r_m\beta^2 + (x_1 - y_1)M}{r_m}\right| = \frac{r_g}{r_m}}$$

Prandtl-Glauert transformation

$$\beta^2 = 1 - M^2$$

$$r_g = \sqrt{(x_1 - y_1)^2 + \beta^2\left((x_2 - y_2)^2 + (x_3 - y_3)^2\right)}$$

- Using the previous result and the property of Dirac delta function:

$$r_m = |\mathbf{x} - \mathbf{y} - \mathbf{U}^{(\infty)}(t - \tau)|$$
$$r_g = r_m\beta^2 + (x_1 - y_1)M$$
$$\left|\frac{\partial g}{\partial \tau}\right| = \frac{r_g}{r_m}$$

$$\Rightarrow \quad \delta(g(\tau)) = \frac{\delta(\tau - \tau_i^*)}{|\partial g/\partial \tau|} = \frac{r_m}{r_g}\delta(\tau - \tau^*)$$

- Green's function for acoustic propagation in a uniform flow:

$$G_e(\mathbf{x}, t|\mathbf{y}, \tau) = \frac{\delta(t - \tau - r_m/c_\infty)}{4\pi r_m} = \frac{\delta(g(\tau))}{4\pi r_m} = \frac{\delta(\tau - \tau^*)}{4\pi r_g}$$

$\tau - \tau^* = 0$ is *equivalent* to $t - \tau - \dfrac{r_g - (x_1 - y_1)M}{c_\infty \beta^2} = 0$

$$\Rightarrow \quad \boxed{G_e(\mathbf{x}, t|\mathbf{y}, \tau) = \frac{\delta\left(t - \tau - \dfrac{r_g - (x_1 - y_1)M}{c_\infty \beta^2}\right)}{4\pi r_g}}$$

Prandtl-Glauert transformation

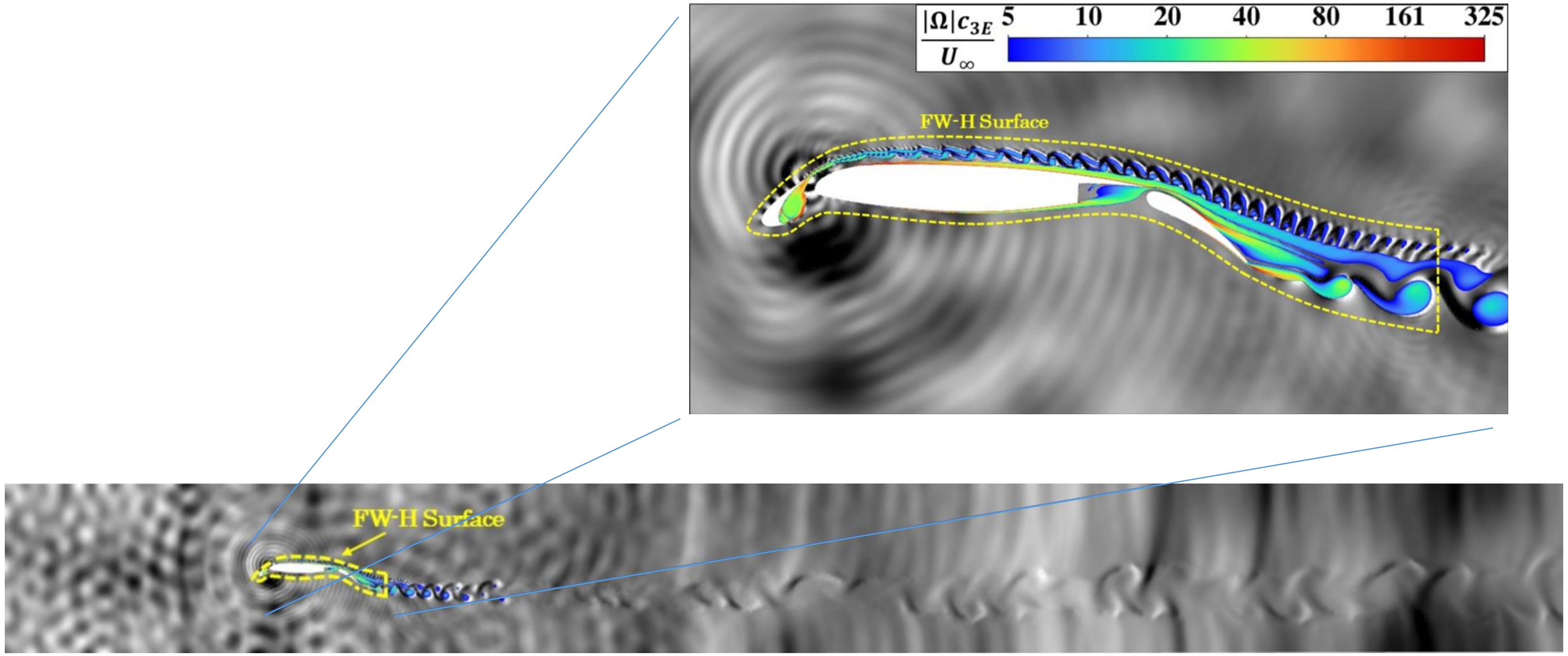
Wave field in a moving frame is the same as a wave field in a stationary frame with the modified space/time variables

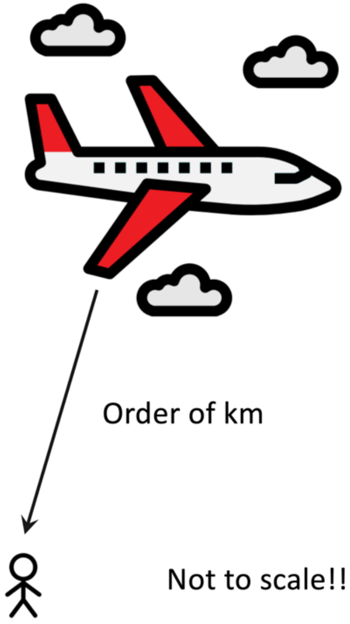
$\xi = (x_1, \beta x_2, \beta x_3)$ and $t_g = t + \dfrac{M x_1}{c_\infty \beta^2}$

- Propagation time depends on $x_1 - y_1$
  - Observer is *upstream* of the source ($x_1 < y_1$) → propagation time *increased* ↑
  - Observer is *downstream* of the source ($x_1 > y_1$) → propagation time *reduced* ↓
- **Amplitude decay rate** - reduced by a factor of r_g/r_m
  - *Stretching* of the waves in the flow direction causes a *contraction* of scales in the directions normal to the mean flow

FW-H surfaces – powerful numerical tool

- FW-H equation can be very powerful for calculating the acoustic far field from CFD simulations
- Impossible to perform a full CFD simulation over a large domain extending to the far-field!
- **FW-H surface** - provide a far-field solution to the wave equation, given *accurate* numerical calculations on a surface that bounds the source region, capturing both **compressible flow fluctuations** and **acoustic waves**





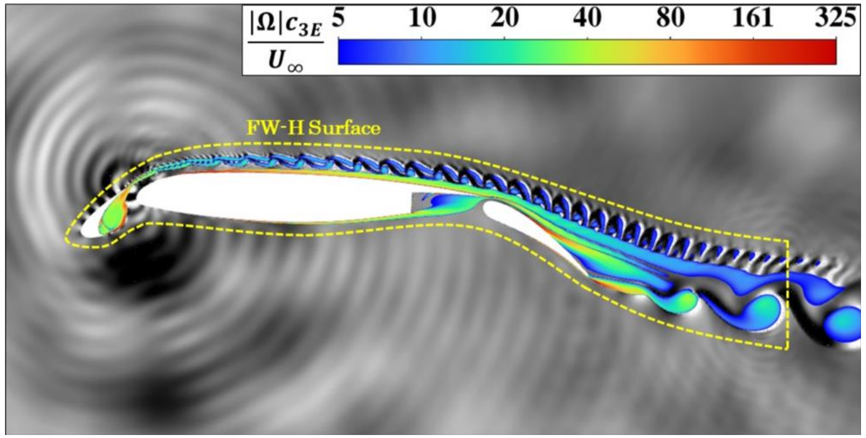
FW-H surface

- FWH surface may be **arbitrarily** located within the numerical domain
  - Highly important because the numerical calculations at the edges of the computational domain may be adversely influenced by BCs
- Commonly, the FW-H surface is placed inside the numerical domain in a region where there is a large *confidence* in the calculations

Choose the FW-H surface so that T_{ij} **does not** contribute to the far-field in the region outside the surface

T_{ij} can be *neglected*

$$\rho'(\mathbf{x},t)c_\infty^2 H_s = -\frac{\partial}{\partial x_i}\int_{S_o}\left[\frac{\left(\rho v_i\left(v_j - V_j\right) + p_{ij}\right)n_j}{4\pi r|1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z}) + \frac{\partial}{\partial t}\int_{S_o}\left[\frac{\left(\rho v_j - \rho' V_j\right)n_j}{4\pi r|1-M_r|}\right]_{\tau=\tau^*} dS(\mathbf{z})$$



FW-H surface

❑ Example (surface decomposition)

$$\langle \hat{P}\hat{P}^* \rangle_{\mathrm{Full}} = \langle \hat{P}\hat{P}^* \rangle_{\mathrm{Airfoil}} + \langle \hat{P}\hat{P}^* \rangle_{\mathrm{Tip}} + 2\Re\left\{ \langle \hat{P}_{\mathrm{Airfoil}} \hat{P}_{\mathrm{Tip}} \rangle \right\}$$

$$\langle \hat{P}\hat{P}^* \rangle_{\mathrm{Tip}} = \langle \hat{P}\hat{P}^* \rangle_{\mathrm{LE}} + \langle \hat{P}\hat{P}^* \rangle_{\mathrm{Mid}} + \langle \hat{P}\hat{P}^* \rangle_{\mathrm{TE}} + 2\Re\left\{ \langle \hat{P}_{\mathrm{LE}} \hat{P}_{\mathrm{Mid}} \rangle + \langle \hat{P}_{\mathrm{Mid}} \hat{P}_{\mathrm{TE}} \rangle + \langle \hat{P}_{\mathrm{LE}} \hat{P}_{\mathrm{TE}} \rangle \right\}$$

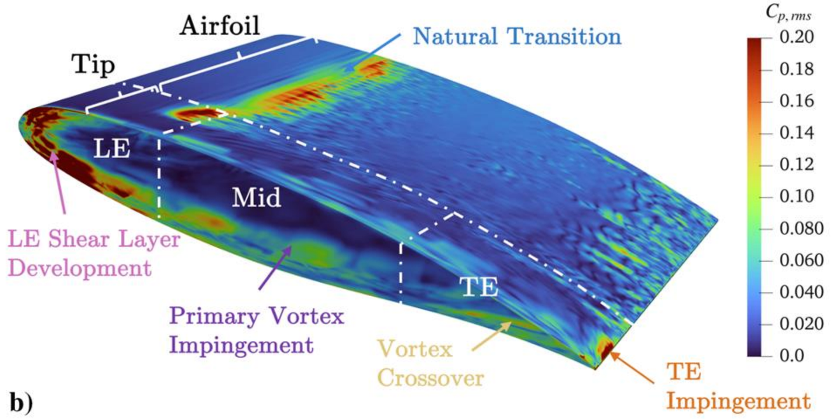


Fig. 11 **RMS pressure coefficient contour on the airfoil surface with acoustic decomposition zones**

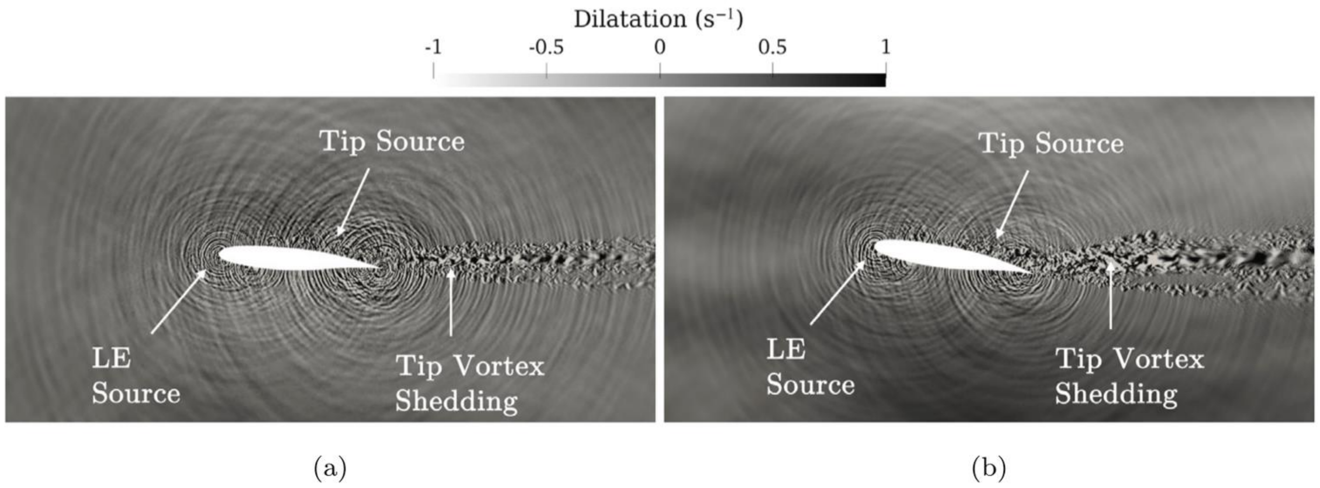


Fig. 18. Spanwise dilatation field at $z/c = 0.02$ from tip: (a) $\alpha = 5°$, (b) $\alpha = 10°$.

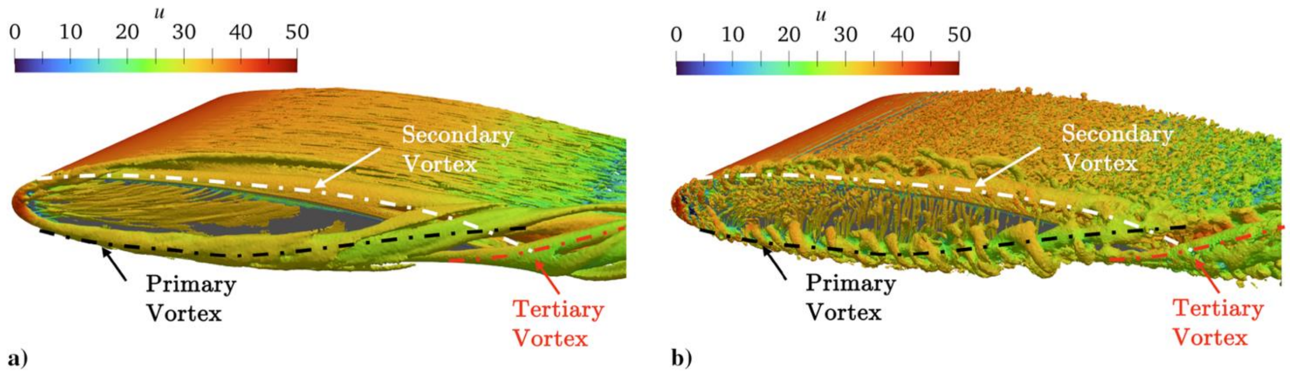


Fig. 6 **Isocontour of the Q-criterion (2×10^5 s^{-2}), colored by the mean streamwise velocity: a) mean isocontour averaged over 3.4 flow-through times, and b) instantaneous isocontour.**

[1] Deng, G. C. *et al.* (2024) "Large-eddy simulation and aeroacoustic prediction of supercritical airfoil side-edge noise", *AIAA Journal* 62(9), pp. 3340-3353.
[2] Deng, G. C. *et al.* (2026) "Influence of angle of attack on airfoil tip noise", *Journal of Sound and Vibration* 620, p. 119376.

FW-H surface

❑ Example (surface decomposition)

$$\langle \hat{P}\hat{P}^* \rangle_{\text{Full}} = \langle \hat{P}\hat{P}^* \rangle_{\text{Airfoil}} + \langle \hat{P}\hat{P}^* \rangle_{\text{Tip}} + 2\Re\left\{ \langle \hat{P}_{\text{Airfoil}} \hat{P}_{\text{Tip}} \rangle \right\}$$

$$\begin{aligned} \langle \hat{P}\hat{P}^* \rangle_{\text{Tip}} &= \langle \hat{P}\hat{P}^* \rangle_{\text{LE}} + \langle \hat{P}\hat{P}^* \rangle_{\text{Mid}} + \langle \hat{P}\hat{P}^* \rangle_{\text{TE}} \\ &\quad + 2\Re\left\{ \langle \hat{P}_{\text{LE}} \hat{P}_{\text{Mid}} \rangle + \langle \hat{P}_{\text{Mid}} \hat{P}_{\text{TE}} \rangle + \langle \hat{P}_{\text{LE}} \hat{P}_{\text{TE}} \rangle \right\} \end{aligned}$$

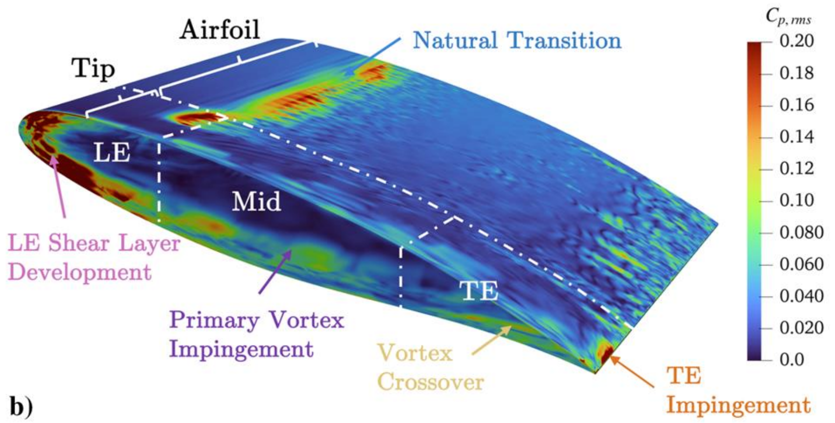


Fig. 11 RMS pressure coefficient contour on the airfoil surface with acoustic decomposition zones

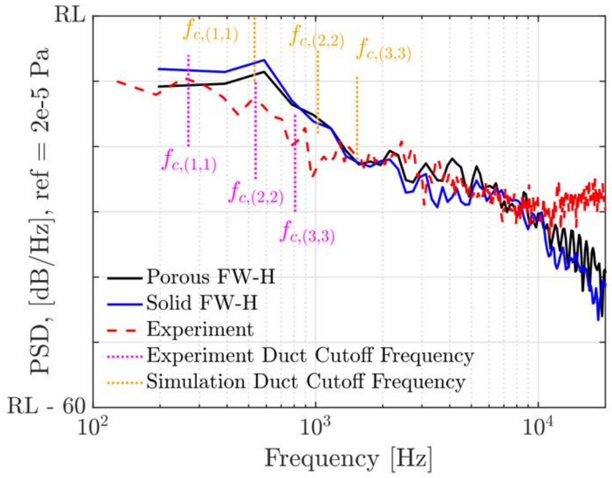


Fig. 20 Far-field noise spectra comparison between experiment, porous FW-H, and solid FW-H data shown in relative levels (RL) for an observer at 1.5 m from the midchord. (Distance between 2 tick marks on PSD represents 10 dB.)

ΔPSD < 2 dB
negligible contribution from quadrupole

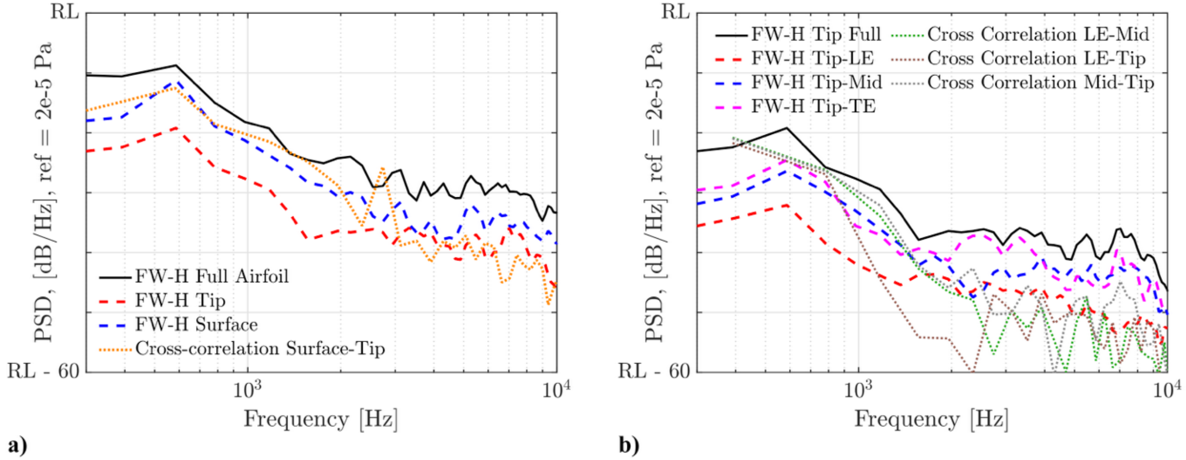
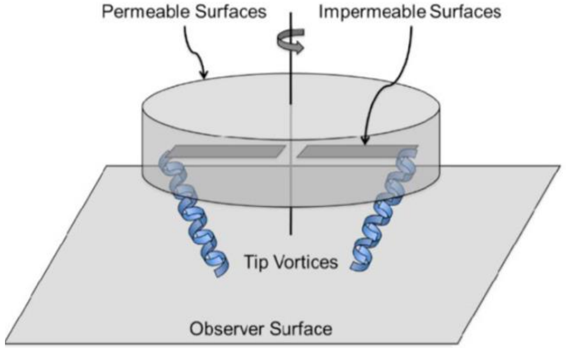


Fig. 21 Acoustic decomposition of far-field noise PSD for different airfoil regions: a) airfoil surface zone and tip zone, and b) LE, Mid, and TE zones. (Distance between 2 tick marks on PSD represents 10 dB.)

Farassat's formulation – F1A

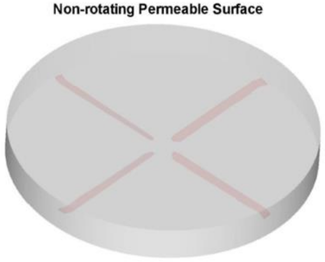
- FW-H surface can also be used for **moving sources** – rotors/propellers
  - One has to calculate the integrand at the emission time for each element of the FWH surface
  - It is advantageous to move the time derivatives inside the integrand

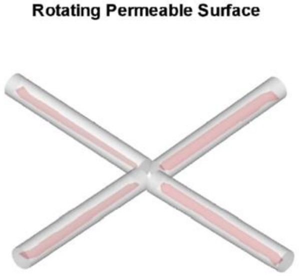


$$\frac{\partial}{\partial x_i} \approx -\frac{x_i}{|\mathbf{x}|}\frac{1}{c_\infty}\frac{\partial}{\partial t} = -\frac{x_i}{|\mathbf{x}|}\frac{1}{c_\infty}\frac{1}{|1-M_r|}\frac{\partial}{\partial \tau}$$

F1A formulation

$$\rho'(\mathbf{x},t)c_\infty^2 H_s \approx \frac{1}{c_\infty}\int_{S_o}\left[\frac{x_i}{4\pi|\mathbf{x}|^2(1-M_r)^2}\left\{\frac{\partial L_i}{\partial \tau}+\frac{L_i}{(1-M_r)}\frac{\partial M_r}{\partial \tau}\right\}\right]_{\tau=\tau^*} dS + \int_{S_o}\left[\frac{1}{4\pi|\mathbf{x}|(1-M_r)^2}\left\{\frac{\partial q}{\partial \tau}+\frac{q}{(1-M_r)}\frac{\partial M_r}{\partial \tau}\right\}\right]_{\tau=\tau^*} dS$$





$$L_i = \left(\rho v_i (v_j - V_j) + p_{ij}\right) n_j$$

$$q = \left(\rho v_j - \rho' V_j\right) n_j$$

Farassat F. (1981) "Linear acoustic formulas for calculation of rotating blade noise", *AIAA Journal* 19, 1122–1130.

Numerical algorithm

Brès et al. (2024), *J. Sound Vib.*

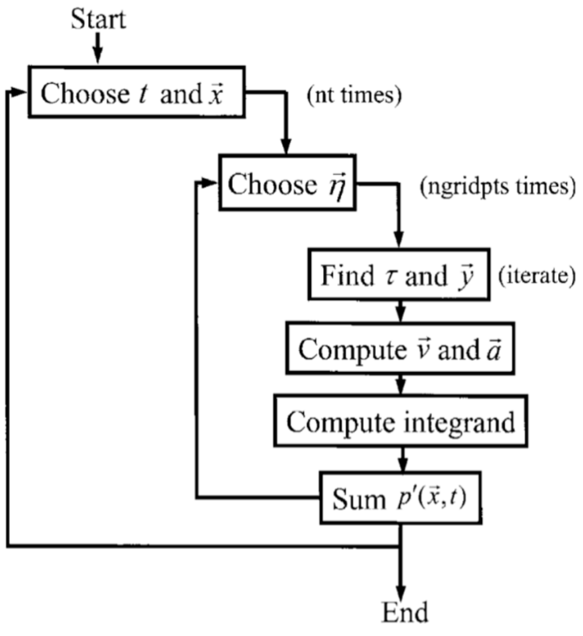


Fig. 1. Schematic of retarded-time algorithm.

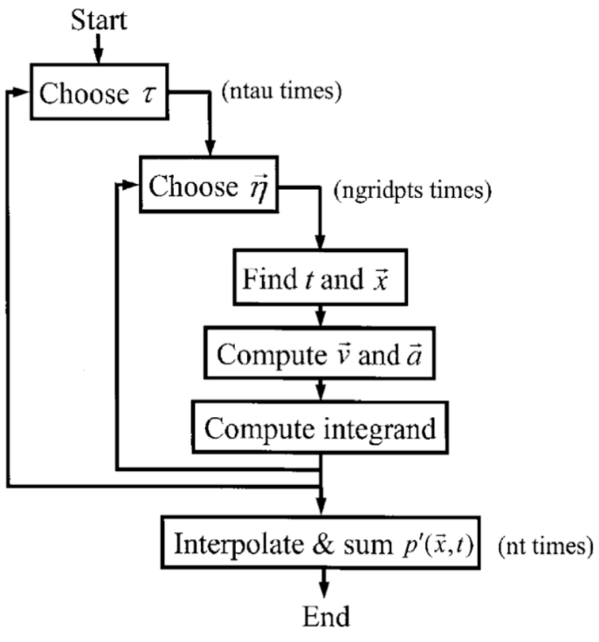
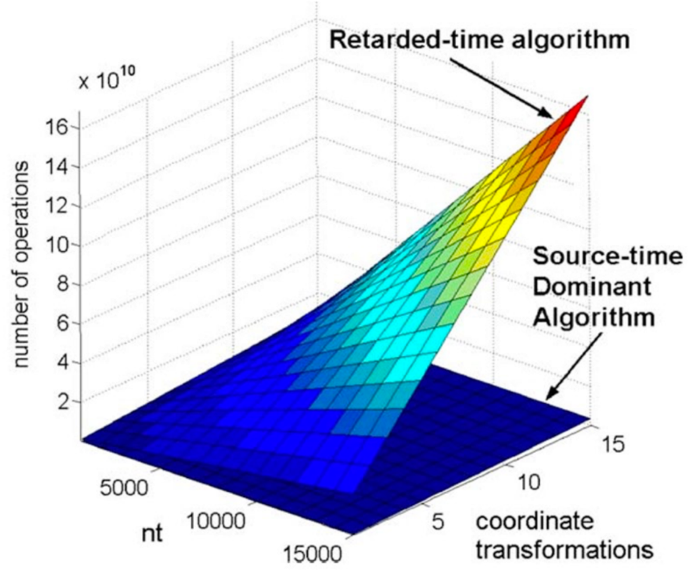


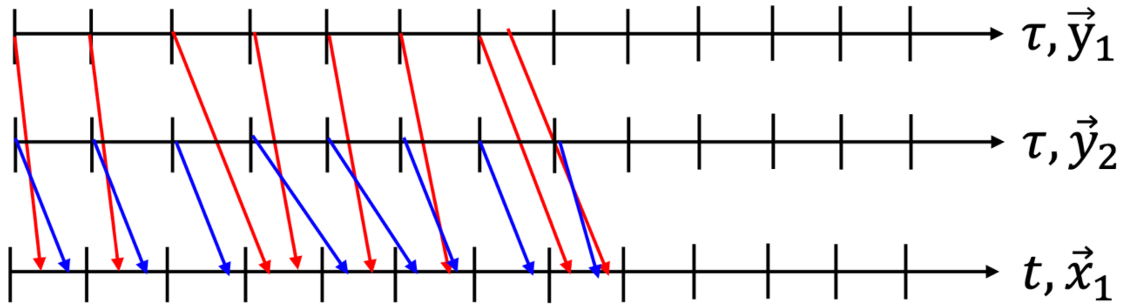
Fig. 2. Schematic of source-time-dominant algorithm.



Source-time-dominant algorithm is much more efficient

Numerical algorithm

- At each source time at each source location, the corresponding observer time is found at each observer.
- This can be repeated for all the source locations over the entire source time range
- Finally, the pressure is interpolated and summed at the designated observer time step
- This entire loop is repeated for each observer



$p'(\vec{x}, t_n)$ is interpolated and summed at the designated observer time step